Tao Wang

Doctoral Candidate
Nanjing University of Aeronautics and Astronautics, China
wangtao21@nuaa.edu.cn • taow1776@gmail.com • daizigege.github.io • Google Scholar Link

Education

- **Ph.D.**, Cyberspace Security, Nanjing University of Aeronautics and Astronautics 2024.04–2027.04
- **Visiting Ph.D.**,Cyberspace Security, University of Milan, SPDP Lab 2025.09–2026.09
- **M.S.**, Cyberspace Security, Nanjing University of Aeronautics and Astronautics 2021.09–2024.04
- **B.E.**, Computer Science and Technology, Anhui Normal University 2017.09–2021.06

Research Interests

- **Field**: Data privacy, computational social science, and artificial intelligence
- **Focus**: Implicit privacy, usable privacy, visual privacy, and privacy evaluation

Previous Research: Controllable Identity Visibility in Usable Face Privacy:
- Universal Privacy Protection: An information-theoretic face anonymization model that removes identity information from both human and machine observers [C1].
- Socially-Aware Privacy: A de-identification approach that preserves recognizability for familiar individuals while preventing machine recognition [J10].
- Human–AI Privacy Asymmetry: An identity hider that leverages differences between human and machine perception, only enabling machine utility [J11].
- Authorized Identity Access: A cancelable face generation model that supports identity recognition exclusively by authorized systems [C3].

Current and Future Research: Human-Centered Implicit Privacy Governance in the GenAI Era.
- As GenAI systems like GPT increasingly infer sensitive attributes, intentions, and social relationships from seemingly innocuous interactions, implicit privacy is emerging as a challenge beyond data disclosure and access control. Such risks arise when AI systems derive sensitive knowledge or social inferences that individuals neither intentionally reveal nor fully anticipate.
- Our research integrates privacy engineering with computational social science to study human-centered implicit privacy. We develop frameworks for implicit privacy discovery, risk awareness, and privacy-preserving interventions, while examining how AI-driven inference shapes disclosure behaviors, privacy perceptions, social interactions, and evolving privacy norms in inference-rich digital environments.

Publications

- **Statistics**: Google scholar citations > 555, h-index = 11; published 7 papers as first or corresponding author, including 4 in CCF-A, 1 with impact factor > 10, and 1 highly cited in ESI

• **Representative publications**: [J14], [J5], and [J3] in IEEE Transactions on Dependable and Secure Computing; [J9] and [J21] in IEEE Transactions on Information Forensics and Security ; [J17] in ACM Computing Surveys; [J7] in IEEE Transactions on Pattern Analysis and Machine Intelligence; [C2–C4] in ACM International Conference on Multimedia

Refereed Journal Papers

J1. Tu, X., Wen, W., **Wang, Tao**, Yang, Z. & Zhang, Y. A Personalized Privacy Protection Service for Cloud-stored Portrait Albums. *IEEE Transactions on Multimedia*, Just Accepted (2026).

J2. **Wang, Tao**, Tao, X., Wen, W., Zhang, Y., Sara, F. & Samarati, P. UnseenGuard: Safeguarding Implicit Visual Privacy with Large Vision-Language Reasoning Graphs. *IEEE Transactions on Multimedia*, Major Revision (2026).

J3. Wen, W., Wu, Z., Zhang, Y., **Wang, Tao**, Xiao, X. & Fang, Y. Medical Archive in an Image: Generating a High-Capacity Customizable Cover Image for Medical Privacy Protection. *IEEE Transactions on Dependable and Secure Computing* **23,** 4735–4747 (2026).

J4. Wen, W., Yuan, Z., Zhang, Y., **Wang, Tao**, Xiao, X., Zhao, R. & Fang, Y. Image Privacy Protection: A Survey. *ACM Computing Surveys* **58,** 1–36 (2026).

J5. Xu, K., Wen, W., Qi, S., **Wang, Tao**, Zhang, Y. & Fang, Y. Detecting Malicious Concepts without Image Generation in AI-Generated Content (AIGC). *IEEE Transactions on Dependable and Secure Computing* **23,** 8699–8716 (2026).

J6. Zhang, X., Fu, Y., **Wang, Tao**, Jiang, Z. & Zhang, M. Mobile-SST-RIR: Resource-aware face image privacy processing for IoT visual devices with key-conditioned reversible recovery. *Internet of Things* **39,** 102061 (2026).

J7. Zhang, Y., Ji, J., **Wang, Tao**, Zhao, R., Wen, W. & Xiang, Y. Make Identity Indistinguishable: Utility-Preserving Face Dataset Publication With Provable Privacy Guarantees. *IEEE Transactions on Pattern Analysis and Machine Intelligence* **48,** 127–139 (2026).

J8. Li, M., Zhou, W., **Wang, Tao**, Zhang, Y. & Wen, W. A Comprehensive Image Protection Framework Based on High-Capacity Adversarial Data Hiding. *IEEE Transactions on Big Data* **11,** 3314–3329 (2025).

J9. **Wang, Tao**, Wen, W., Xiao, X., Hua, Z., Zhang, Y. & Fang, Y. Beyond Privacy: Generating Privacy-Preserving Faces Supporting Robust Image Authentication. *IEEE Transactions on Information Forensics and Security* **20,** 2564–2576 (2025).

J10. **Wang, Tao**, Zhang, Y., Xiao, X., Xu, K., Yuan, L., Wen, W. & Fang, Y. Make Identity Unextractable yet Perceptible: Synthesis-Based Privacy Protection for Subject Faces in Photos. *arXiv preprint arXiv:2509.11249* (2025).

J11. **Wang, Tao**, Zhang, Y., Yang, Z., Xiao, X., Zhang, H. & Hua, Z. Seeing is Not Believing: An Identity Hider for Human Vision Privacy Protection. *IEEE Transactions on Biometrics, Behavior, and Identity Science* **7,** 170–181 (2025).

J12. Xie, Y., Zhou, Y., **Wang, Tao**, Hua, Z., Wen, W., Yi, S. & Zhang, Y. Controllable facial protection against malicious translation-based attribute editing. *Knowledge-Based Systems* **309,** 112873 (2025).

J13. Zhang, X., **Wang, Tao**, Ji, J., Zhang, Y. & Lan, R. Privacy-preserving face attribute classification via differential privacy. *Neurocomputing* **626,** 129556 (2025).

J14. Zhang, Y., Yang, Z., **Wang, Tao**†, Hua, Z. & Weng, J. Tailor-Made Face Privacy Protection via Class-Wise Targeted Universal Adversarial Perturbations. *IEEE Transactions on Dependable and Secure Computing* **22,** 5108–5120 (2025).

J15. Zhao, R., Zhang, Y., **Wang, Tao**, Wen, W., Xiang, Y. & Cao, X. Visual content privacy protection: A survey. *ACM Computing Surveys* **57,** 1–36 (2025).

J16. Ji, J., Zhao, R., **Wang, Tao** & Zhu, Y. Utility-enhanced image obfuscation with block differential privacy. *IEEE Signal Processing Letters* **31,** 2135–2139 (2024).

J17. **Wang, Tao**, Zhang, Y., Qi, S., Zhao, R., Xia, Z. & Weng, J. Security and Privacy on Generative Data in AIGC: A Survey. *ACM Computing Surveys* **57,** 1–34 (2024).

J18. **Wang, Tao**, Zhang, Y., Zhao, R., Wen, W. & Zhu, Y. Protecting Face Privacy via Beautification. *Journal of Frontiers of Computer Science & Technology* **18,** 244 (2024).

J19. Xie, Y., Zhou, Y., **Wang, Tao**†, Wen, W., Yi, S. & Zhang, Y. Reversible Gender Privacy Enhancement via Adversarial Perturbations. *Neural Networks* **172,** 106130 (2024).

J20. **Wang, Tao**, Zhang, Y., Zhao, R., Wen, W. & Lan, R. Identifiable face privacy protection via virtual identity transformation. *IEEE Signal Processing Letters* **30,** 773–777 (2023).

J21. Zhang, Y., **Wang, Tao**, Zhao, R., Wen, W. & Zhu, Y. RAPP: Reversible Privacy Preservation for Various Face Attributes. *IEEE Transactions on Information Forensics and Security* **18,** 3074–3087 (2023).

J22. Sun, L., Liu, X., Liu, Y., **Wang, Tao**, Guo, L., Zheng, X. & Luo, Y. A novel deep recommend model based on rating matrix and item attributes. *Journal of Intelligent Information Systems* **57,** 295–319 (2021).

Refereed Conference Papers

C1. **Wang, Tao**. *Information-Theoretic Face Anonymization* in *Under Review* (2026).

C2. Xu, K., Zhang, Y., **Wang, Tao**, Qi, S., Carminati, B., Ferrari, E. & Fang, Y. *Concept-Level Risk and Calibration for Governance in Diffusion Foundation Models* in *Proceedings of the ACM International Conference on Multimedia* (2026), Just Accepted.

C3. **Wang, Tao**, Zhang, Y., Xiao, X., Yuan, L., Xia, Z. & Weng, J. *Make Privacy Renewable! Generating Privacy-Preserving Faces Supporting Cancelable Biometric Recognition* in *Proceedings of the ACM International Conference on Multimedia* (2024), 10268–10276.

C4. Yang, Z., Zhang, Y., **Wang, Tao**, Hua, Z., Xia, Z. & Weng, J. *Once-for-all: Efficient Visual Face Privacy Protection via Person-Specific Veils* in *Proceedings of the ACM International Conference on Multimedia* (2024), 7705–7713.

Awards & Honors

- Doctoral Student Program of the Young Sci-Tech Talents Cultivation Project, CAST, 2025
- Puxin Elite Scholarship, 2025
- CSC Scholarship of High-Level Postgraduate Program, 2025
- ACM MM Honorable Mention Award, 2024
- National Scholarships for Postgraduate Students, 2023

Research Grants

• **PI**, Postgraduate Research Innovation Program, Jiangsu Provincial Department of Education; Friendly Face Privacy Protection — CN¥ 15,000 — 2025–2027

Presentations & Talks

2026 MILAN0DAY, University of Milan. Invited Talk: "AI and the Death of Privacy? Navigating New Threats and Modern Defenses."

Professional Services

Journal Reviewers: TKDE, TDSC, TCSVT, TNSE, TBIOM, TOMM, EAAI, ESWA, KBS, IOTJ.
Conference Reviewer: ICLR, SIGKDD, AAAI, ACM MM
Membership: ACM EIGTRUST Membership; CSIG Membership

References

• **Pierangela Samarati**, General supervisor, pierangela.samarati@unimi.it , University of Milan
• **Yushu Zhang**, Ph.D. supervisor, zhangyushu@jxufe.edu.cn, Jiangxi University of Finance and Economics
• **Yuming Fang**, General supervisor, leo.fangyuming@foxmail.com, , Jiangxi University of Finance and Economics
• **Shuren Qi**, General supervisor, shurenqi@nuaa.edu.cn, City University of Hong Kong

---

Last updated: October 24, 2026